

The phase-winding bit: how much noise does a topological invariant actually survive?

A quantitative note, with a correction of one common misapplication

O. Kirichenko

urevich55@gmail.com

July 4, 2026

Abstract

“Information stored in a topological invariant is protected against continuous deformation” is a true and powerful statement—it underlies topological quantum error correction and won recognition in the 2016 Nobel Prize for topological phases of matter. It is also frequently *misapplied*, e.g. to claim that a macroscopic object could be “dissolved and reconstructed” because its structure is topologically encoded. This note does two things. First, it makes the true statement *quantitative* for the simplest topological memory—the winding number of a phase field—by Monte-Carlo: the winding survives local Gaussian phase noise up to a sharp threshold ($\sigma \approx 0.6$ rad here), below which the bit-error rate falls exponentially as $\sim N \operatorname{erfc}(\pi/2\sigma)$ (Fig. 1). Second, it draws the honest boundary: the same integer winding that robustly stores *one bit* (and that appears, correctly, as the topological charge in the arrow-of-time argument of the NVG framework) cannot store the $\sim 10^{26}$ independent coordinates of a macroscopic body—the protection is real, but its information capacity is $\log_2$ of the number of topological sectors, not the phase-space volume.

1 The claim, and where it goes wrong

A phase field $\theta(x) \in [0, 2\pi)$ on a ring carries a winding number

$$W = \frac{1}{2\pi} \oint \partial_x \theta dx = \frac{1}{2\pi} \sum_i \operatorname{wrap}(\theta_{i+1} - \theta_i) \in \mathbb{Z}, \quad (1)$$

where wrap maps a difference into $(-\pi, \pi]$. W is an *integer*: no continuous, local deformation of θ can change it, because to do so some link difference must cross $\pm\pi$ —a discrete “phase slip”—which a smooth deformation never forces. This is the prototype of topological protection: the stored datum (W) is a global property, invisible to and unchangeable by local perturbations, until they are large enough to tear the field.

The correct uses of this fact are among the deepest in modern physics: Kitaev’s toric code and the surface codes store logical qubits in the homology (winding) sectors of a lattice, protected against local errors up to a threshold [1, 2]; the 2016 Nobel Prize recognised topological order and the role of winding invariants in matter [3]. Within the NVG/VMF framework the *same* integer appears legitimately as the topological charge $Q = \frac{1}{2\pi} \oint d\theta = 1$ that fixes the sign of entropy production in the arrow-of-time argument.

The *misapplication* is to read “topologically protected” as “arbitrarily much information, indestructibly stored.” A winding number protects exactly $\log_2(\text{number of accessible sectors})$ bits—one bit per independent $\mathbb{Z}$ (or $\log_2 d$ for a d -sector code), not the $\sim 10^{26}$ real numbers specifying the atomic coordinates of a kilogram of matter. Topology protects the *label*, not the *library*.

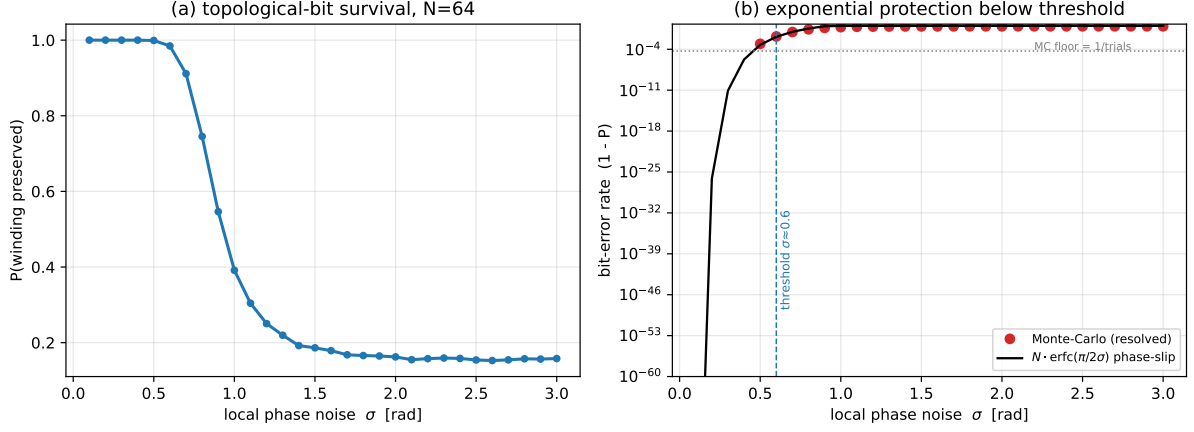


Figure 1: (a) Probability that the winding number is preserved vs local phase-noise amplitude σ ($N = 64$, 2×10^4 Monte-Carlo trials). The bit is essentially perfectly protected below $\sigma \approx 0.6$ rad and randomises above $\sigma \sim 1$ rad. (b) Bit-error rate: Monte-Carlo (points, resolution-limited to $1/\text{trials}$) against the analytic phase-slip estimate $N \text{erfc}(\pi/2\sigma)$ (line), which falls exponentially below threshold—the signature of topological protection. Reproduced by `winding_bit.py`.

2 How much noise does one winding bit survive?

We store $W = 1$ as the smooth state $\theta_i = 2\pi i/N$ on a ring of $N = 64$ sites, add independent Gaussian noise $\theta_i \rightarrow \theta_i + \mathcal{N}(0, \sigma)$, and measure the probability that the winding is still 1. Figure 1a shows a sharp survival transition; Fig. 1b shows the bit-error rate.

Two quantitative lessons:

- **A threshold, not a wall.** Below $\sigma_c \approx 0.6$ rad the error is exponentially small— $N \text{erfc}(\pi/2\sigma)$, e.g. $\sim 10^{-23}$ at $\sigma = 0.3$ for $N = 64$ —so the bit is stable for cosmological times. Above threshold it randomises. Protection is parametric, controlled by the ratio (noise)/(tear scale π), exactly as topological quantum codes have an error threshold.
- **Capacity scales, robustness dilutes.** The error grows linearly with system size at fixed σ (0.34, 0.60, 0.82 for $N = 16, 64, 256$): more links means more chances for a phase slip. A larger medium stores *more* winding sectors (more bits) but each individual bit is *easier* to break—the standard capacity/robustness trade that real topological codes resolve with redundancy and active correction.

3 The honest boundary

The winding bit is the correct, minimal realisation of the intuition “structure encoded in a topological invariant resists continuous deformation.” Its value—and its limit—are now numbers:

1. it stores $\log_2(\#\text{sectors})$ bits, robustly, with an explicit noise threshold;
2. it does *not* store a macroscopic configuration: topology protects a discrete label whose capacity is logarithmic in the number of sectors, astronomically short of the phase-space of a solid body;
3. the same invariant is used correctly elsewhere—toric/surface codes for qubits, and the topological charge $Q = 1$ in the NVG arrow-of-time argument—while the “teleport a body by topological encoding” reading fails on the capacity count alone, before any dynamical objection.

The general moral is a useful filter: whenever “topologically protected” is invoked, ask *how many bits* the invariant actually carries and *what tears it*. Both are computable, as above; a claim that survives both is real physics, and one that needs the invariant to carry macroscopic information is not.

Code and data

`winding_bit.py` reproduces the figure, the threshold and the N -scaling (ring phase field, exact integer winding, Monte-Carlo over local Gaussian noise).

References

- [1] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Ann. Phys.* **303** (2003) 2 [quant-ph/9707021].
- [2] E. Dennis, A. Kitaev, A. Landahl, J. Preskill, “Topological quantum memory,” *J. Math. Phys.* **43** (2002) 4452.
- [3] The Nobel Prize in Physics 2016 (D. J. Thouless, F. D. M. Haldane, J. M. Kosterlitz), “for theoretical discoveries of topological phase transitions and topological phases of matter.”